

Dr Mélodie Monod

CONTACT DETAILS	Email: melodie.monod@dauphine.psl.eu Website: https://melodiemonod.github.io/	
EDUCATION	Imperial College London , London, United Kingdom	Oct 2019 - Dec 2022
	<i>PhD in Modern Statistics and Statistical Machine Learning</i> Fully funded by EPSRC. First student to complete the program among all cohorts. Supervisors: Dr. Oliver Ratmann, Prof. Samir Bhatt Thesis title: <i>Bayesian models and methods to estimate age-specific infectious disease transmission dynamics: integrating disease surveillance time series, mobility, and vaccination data.</i>	
	Imperial College London , London, United Kingdom	Sep 2018 - Sep 2019
	<i>Master of Science in Statistics</i> Final Grade: Distinction.	
	University of Geneva , Geneva, Switzerland	Sep 2015 - Jun 2018
	<i>Bachelor of Science in Economics and Management</i> Major in Economics, Minor in Statistics. Final Grade: 5.71/6, Class Rank: First out of > 100 students.	
EXPERIENCE	Université Paris Dauphine – PSL , Paris, France	Since
	October 2025 Senior Fellow Conducting research in Bayesian probabilistic modeling, exploring the interplay between deep learning and Bayesian inference to develop robust, interpretable, and data-efficient probabilistic models.	
	University of Oxford , Oxford, United Kingdom	June 2025 - September 2025
	<i>Senior Postdoctoral Researcher</i> Conducted research on Deep Generative Modelling.	
	Imperial College London , London, United Kingdom	Oct 2024 - May 2025
	<i>Research Associate (Postdoctoral Researcher)</i> Conducted research on Bayesian deep learning and diffusion models.	
	Novartis , Dublin, Ireland	Jan 2023 - Sep 2024
PROGRAMMING SKILLS	<i>Principal Biostatistician</i> Developed original statistical methods for pharmaceutical applications, specializing in deep causal analysis and deep survival analysis.	
	Imperial College London , London, United Kingdom	Apr 2023 - Sep 2024
	<i>Honorary Research Associate</i> Engaged in research collaborations with academic faculty, PhD, and MSc students on projects involving Bayesian statistics, phylodynamics, and mathematical models of infectious disease dynamics alongside Dr. Oliver Ratmann.	
	Novartis , Basel, Switzerland	Jun 2021 - Sep 2021
	<i>Advanced Explanatory Analysis Intern</i> Supervisor: Dr. Sebastian Weber Developed survival analysis methods for assessing safety events in phase I oncology trials accounting for incomplete PK data.	
PROGRAMMING SKILLS	Highly proficient in Python, R and Stan.	
	Open source code: • NeuralSurv: Deep Survival Analysis with Bayesian Uncertainty Quantification associated with [1]	

- [Deep Reinforcement Learning for Online Optimal Execution Strategies](#) associated with [3]
- [TorchSurv](#), a package for deep survival analysis associated with [4]
- [Analysis of age- and time-specific HIV transmission dynamics](#) associated with [5]
- [Regularised B-splines Projected Gaussian Process Priors](#) associated with [6]
- [Model for characterizing COVID-19 spread](#) associated with [8, 9, 10]

SELECTED PUBLICATIONS

- [1] Mélodie Monod, Alessandro Micheli, and Samir Bhatt. “NeuralSurv: Deep Survival Analysis with Bayesian Uncertainty Quantification”. In: *39th Conference on Neural Information Processing Systems (NeurIPS)* (2025).
- [2] Mélodie Monod, Alessandro Micheli, and Samir Bhatt. “Diffusion Models for Inverse Problems in the Exponential Family”. In: *arXiv* (2025). eprint: [arXiv:2502.05994](#).
- [3] Mélodie Monod and Alessandro Micheli. “Deep Reinforcement Learning for Online Optimal Execution Strategies”. In: *arXiv* (2024). eprint: [arXiv:2410.13493](#).
- [4] Mélodie Monod, Peter Krusche, Qian Cao, Berkman Sahiner, Nicholas Petrick, David Ohlssen, and Thibaud Coroller. “TorchSurv: A Lightweight Package for Deep Survival Analysis”. In: *Journal of Open Source Software* 9.104 (2024), p. 7341. DOI: [10.21105/joss.07341](#).
- [5] Mélodie Monod, Andrea Brizzi, Ronald M. Galiwango, Robert Ssekubugu, Yu Chen, Xiaoyue Xi, Edward Nelson Kankaka, Victor Ssempijja, Lucie Abeler-Dörner, Adam Akullian, Alexandra Blenkinsop, David Bonsall, et al. “Longitudinal population-level HIV epidemiologic and genomic surveillance highlights growing gender disparity of HIV transmission in Uganda”. In: *Nature Microbiology* 9.1 (2024), pp. 35–54. ISSN: 2058-5276. DOI: [10.1038/s41564-023-01530-8](#).
- [6] Mélodie Monod, Alexandra Blenkinsop, Andrea Brizzi, Yu Chen, Carlos Cardoso Correia Perello, Vidoushee Jogarah, Yuanrong Wang, Seth Flaxman, Samir Bhatt, and Oliver Ratmann. “Regularised B-splines Projected Gaussian Process Priors to Estimate Time-trends in Age-specific COVID-19 Deaths”. In: *Bayesian Analysis* 18.3 (2023). ISSN: 1936-0975. DOI: [10.1214/22-ba1334](#).
- [7] Nuno R. Faria, Thomas A. Mellan, Charles Whittaker, Ingra M. Claro, Darlan da S. Candido, Swapnil Mishra, Myuki A. E. Crispim, Flavia C. S. Sales, Iwona Hawryluk, John T. McCrone, Ruben J. G. Hulswit, Lucas A. M. Franco, et al. “Genomics and epidemiology of the P.1 SARS-CoV-2 lineage in Manaus, Brazil”. In: *Science* 372.6544 (2021), pp. 815–821. DOI: [10.1126/science.abh2644](#).
- [8] Mélodie Monod, Alexandra Blenkinsop, Xiaoyue Xi, Daniel Hebert, Sivan Bershman, Simon Tietze, Marc Baguelin, Valerie C. Bradley, Yu Chen, Helen Coupland, Sarah Filippi, Jonathan Ish-Horowicz, et al. “Age groups that sustain resurging COVID-19 epidemics in the United States”. In: *Science* 371.6536 (2021). ISSN: 1095-9203. DOI: [10.1126/science.abe8372](#).
- [9] H. Juliette T. Unwin, Swapnil Mishra, Valerie C. Bradley, Axel Gandy, Thomas A. Mellan, Helen Coupland, Jonathan Ish-Horowicz, Michaela A. C. Vollmer, Charles Whittaker, Sarah L. Filippi, Xiaoyue Xi, Mélodie Monod, et al. “State-level tracking of COVID-19 in the United States”. In: *Nature Communications* 11.1 (2020). DOI: [10.1038/s41467-020-19652-6](#).
- [10] Seth Flaxman, Swapnil Mishra, Axel Gandy, H. Juliette T. Unwin, Thomas A. Mellan, Helen Coupland, Charles Whittaker, Harrison Zhu, Tresnia Berah, Jeffrey W. Eaton, Mélodie Monod, Pablo N. Perez-Guzman, et al. “Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe”. In: *Nature* 584.7820 (2020), pp. 257–261. ISSN: 1476-4687. DOI: [10.1038/s41586-020-2405-7](#).

GRANTS	Paris AI Research Institute (PR[AI]RIE-PSAI) Fellowship	Oct 2025
	Four-year research fellowship supporting independent research in machine learning and Bayesian probabilistic modeling.	
	LMS Travel Grant for Early Career Researchers, £500.00	Apr 2025
	EPSRC CDT in Modern Statistics and Statistical Machine Learning Scholarship	
	Imperial College London, London, United Kingdom	2019-2023
	The Department of Mathematics Scholarship for the MSc Statistics	
	Imperial College London, London, United Kingdom	2018
	Awarded for academic excellence and ability to continue into a postgraduate research degree.	
AWARDS	Savage Award	
	Finalist and Honorable Mention for outstanding doctoral dissertation in Bayesian statistics.	2025
	Highest overall grade average of the Bachelor's Degree in Economics and Management - Major Economics	
	University of Geneva, Geneva, Switzerland	2018
TEACHING	Lecturer , PSL University	Fall 2025
	EIB11FA005 Statistics for the International Bachelor of Science in AI.	
	Graduate Teaching Assistant , Imperial College London	Spring 2021
	MATH97134 Biomedical Statistics for postgraduate in the MSc of Statistics.	
	Senior Graduate Teaching Assistant , Imperial College London	Spring 2021
	MATH40006 Introduction to computation for first year maths undergraduates	
	Graduate Teaching Assistant , Imperial College London	Fall 2020
	MATH40006 Introduction to computation for first year maths undergraduates	
TALKS	London Geometry and Machine Learning Summer School, project presentation	2025
	Machine Learning for Global Health	2025
	PyTorch TAC Meeting	2025
	Careers in Statistics, Imperial College London	2024
	Computer Science and Machine Learning group, Imperial College London	2022
	ISBA World Meeting	2022
	Banff International Research Station for Mathematical Innovation and Discovery	2021
	COVID-19 Dynamics & Evolution Webinar Series	2020
	StanCon	2020
SUPERVISION	3 MSc students : 2 supervised and 1 co-supervised. 1 intern in the industry.	
SERVICE TO PROFESSION	Reviewer: Annals of Statistics, BMC Global Health, PLOS Global Public Health.	
	Guest Editor: Mathematics.	
	Seminar organisation: Organiser of the “Generative Modelling” seminar at I-X (Department of AI of Imperial College London) in April 2025.	